

Scripting Language (CACS254)

Unit 1: Client-Side Scripting (JavaScript) — Laboratory Questions

Total Experiments: 30

JavaScript-specific; DOM, event-handling, and validation topics that recur most often in the Semester-End Exam; arranged in logical sequence from beginner to advanced, suitable for lab reports and exam preparation.

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Section A: JavaScript Fundamentals & Embedding

1. Experiment 1: Hello World Program and Embedding JavaScript in HTML

Question: Write a JavaScript program to display the message “Hello World!” on a web page. Demonstrate the three different ways of embedding JavaScript in an HTML document: Inline JavaScript, Internal JavaScript, and External JavaScript.

2. Experiment 2: Variables, Data Types, Comments, Input and Output

Question: Write a JavaScript program demonstrating variables, data types, single-line comments, multi-line comments, user input using `prompt()`, and output using `alert()` and `document.write()`.

3. Experiment 3: JavaScript Operators

Question: Write a JavaScript program to demonstrate arithmetic, assignment, comparison, logical, and ternary operators.

Section B: Decision Making

4. Experiment 4: Largest Among Three Numbers

Question: Write a JavaScript program to find the largest among three numbers entered by the user, using if-else statements.

5. Experiment 5: Day of the Week Using Switch

Question: Write a JavaScript program to display the corresponding day of the week (1–7) using the switch statement.

Section C: Looping Statements

6. Experiment 6: Multiplication Table

Question: Write a JavaScript program to display the multiplication table of a given number.

7. Experiment 7: Fibonacci Series

Question: Write a JavaScript program to generate the Fibonacci series up to N terms.

Section D: Arrays

8. Experiment 8: Array Operations – Sum, Average, Largest, Smallest

Question: Write a JavaScript program to accept n integers into an array and calculate and display the sum, average, largest, and smallest values among the elements.

9. Experiment 9: Student Marks Processing

Question: Write a JavaScript program to store marks of five subjects in an array and calculate the total, percentage, and division.

10.Experiment 10: Countries Table

Question: Write a JavaScript program that stores fifty country names in an array and displays them in an HTML table with four countries in each row.

11.Experiment 11: Array Methods – forEach() and map()

Question: Write a JavaScript program demonstrating the use of forEach() and map() methods with suitable examples.

12.Experiment 12: Reverse First and Last Name

Question: Write a JavaScript program that accepts the user's first name and last name and displays them in reverse order, separated by a space.

Section E: Functions

13.Experiment 13: User-Defined Functions

Question: Write a JavaScript program to perform basic arithmetic operations using user-defined functions.

14.Experiment 14: Arrow Functions

Question: Write a JavaScript program to calculate the factorial and square of a number using Arrow Functions.

15.Experiment 15: Variable-Length Arguments Function

Question: Write a JavaScript function sum() that accepts any number of arguments and returns their total. For example, sum(1, 2) should return 3, and sum(1, 3, 4) should return 8.

16.Experiment 16: Immediately Invoked Function Expression (IIFE)

Question: Write a JavaScript program to demonstrate the use of an Immediately Invoked Function Expression (IIFE) with a suitable example.

Section F: Built-in Objects

17.Experiment 17: Math, String and NaN Objects

Question: Write a JavaScript program demonstrating the use of Math object, String object, and NaN with suitable examples.

18.Experiment 18: Date Object and Live Digital Clock

Question: Write a JavaScript program to display the current date and time and create a live digital clock in HH:MM:SS format.

Section G: Browser Objects

19.Experiment 19: Dialog Boxes

Question: Write a JavaScript program demonstrating the use of `alert()`, `confirm()`, and `prompt()` dialog boxes.

20.Experiment 20: Browser Window Operations

Question: Write a JavaScript program demonstrating browser window operations such as `window.open()`, `window.close()`, and `window.print()`.

Section H: Document Object Model (DOM)

21.Experiment 21: DOM Selectors

Question: Write a JavaScript program demonstrating the use of `getElementById()`, `getElementsByClassName()`, `getElementsByName()`, `querySelector()`, and `querySelectorAll()`.

22.Experiment 22: Dynamic DOM Manipulation

Question: Write a JavaScript program to dynamically create and remove list items of an unordered list using the Document Object Model (DOM).

23.Experiment 23: Highlight Long Words

Question: Write a JavaScript program to highlight all words longer than eight characters (with a background color) in a paragraph.

24.Experiment 24: Image Hover Description

Question: Create an HTML page containing an image and a paragraph. Write a JavaScript program to change the paragraph description during `onmouseover` and `onmouseout` events on the image.

Section I: Event Handling and Integrated Applications

25.Experiment 25: JavaScript Event Handling

Question: Write a JavaScript program demonstrating the use of `onclick`, `ondblclick`, `onmouseover`, `onmouseout`, `onchange`, and `onkeyup` events.

26.Experiment 26: Radio Button and Automatic Background Color Changer

Question: Write a JavaScript program to change the background color of the page based on a radio button selection (green, red, blue). Also write a program to automatically change the background color of the page every second.

27.Experiment 27: Arithmetic Calculator

Question: Design and develop a basic calculator using HTML and JavaScript. When the user presses numbers or an arithmetic operator, it should append to the input display, and when the user presses the equals (=) button, the entered expression should be calculated and displayed.

28.Experiment 28: Interactive Registration Form with Validation

Question: Design and develop a Registration Form using HTML and JavaScript. Validate the following fields using client-side validation and Regular Expressions:

- Name
- Email Address
- Mobile Number
- Password
- Date of Birth

Section J: Cookies**29.Experiment 29: Cookie Basics**

Question: Write a JavaScript program to create, read, and delete a cookie on a web page.

30.Experiment 30: Advanced Cookie Handling

Question: Write a JavaScript program to set a cookie with an expiry date and use it to remember a user's preference (for example, a "Remember Me" option or last-visited time) across page reloads.